4.2 Bootstrap Estimation

- Methodology took its name in a paper by Efron (1979).
- Computer intensive method.
- Mainly distinguished in two types: **parametric** and **non-parametric** bootstrap.

Suppose that we want to make inferences about parameter θ using observed data $y_1, y_2, \ldots, y_n$, which follow a distribution with cdf $F(y;\theta)$.

Usually inference will be based on the **sampling distribution** of an estimate $\hat{\theta}$.

A sampling distribution is obtained either by theoretical results, or is based on many samples from $F(y;\theta)$.

Alternatively, bootstrap constructs samples by **repeatedly sampling with replacement** from the available sample $y_1, y_2, \ldots, y_n$

Bootstrap Estimation (continued)

Basic idea (non-parametric bootstrap)

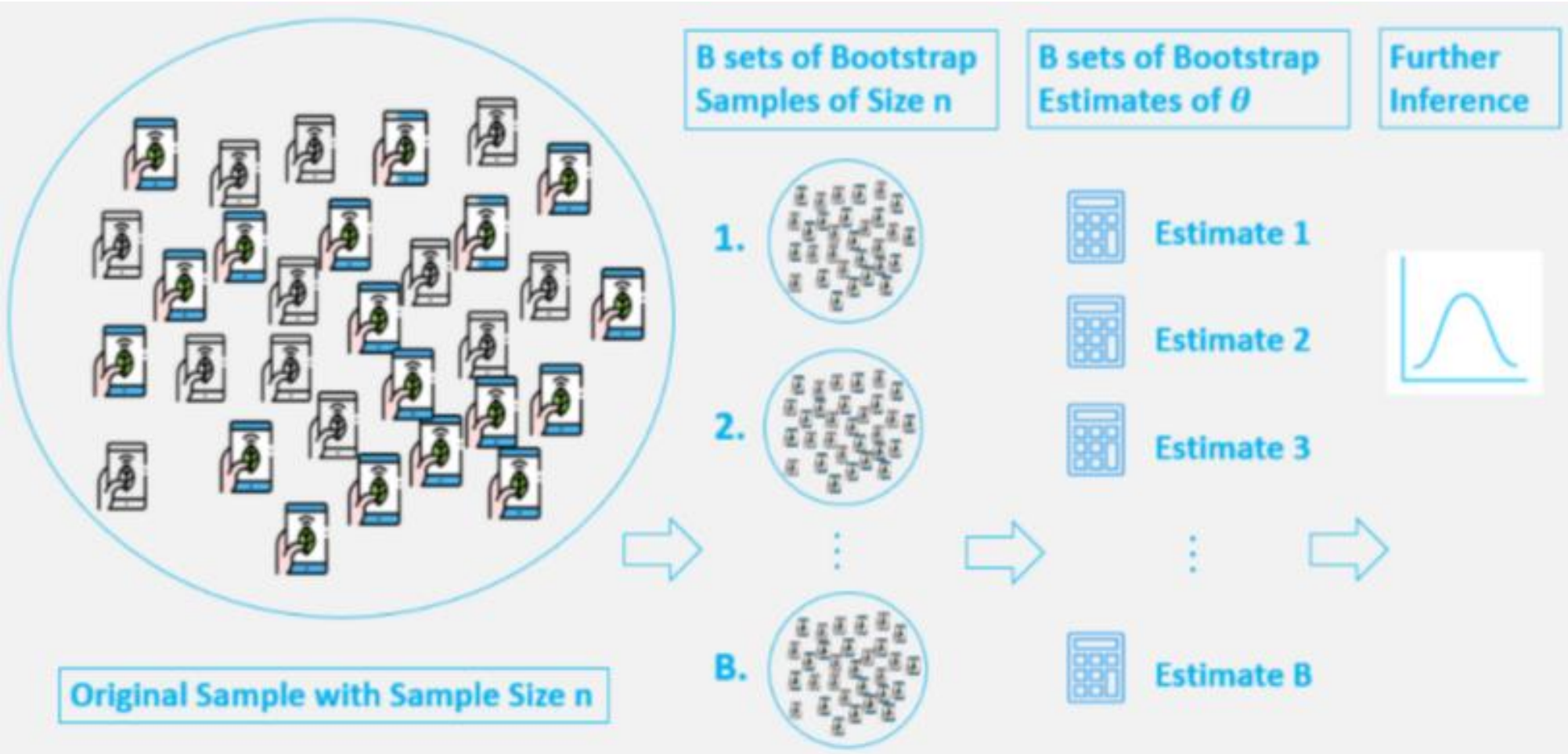
Construct an empirical distribution, $\hat{F}_n$:

$$\hat{F}_n(y) = \frac{1}{n}\{\text{Number of } y_i \leq y\}$$

1. Draw a sample of size n from $\hat{F}$
   This is the bootstrap sample $y_1^*, y_2^*, \ldots, y_n^*$ with y^*s selected **with replacement** from $y_1, y_2, \ldots, y_n$
   This is the *non-parametric (full) bootstrap*.
2. Obtain an estimate $\hat{\theta}^*$ from the bootstrap sample.
   This is done in the same way as $\hat{\theta}$ is obtained from the original sample.
3. Repeat steps 1 and 2(say B times).

Repetition of the above steps will provide the empirical distribution of $\hat{\theta}^*$, which serves as an approximation of the distribution of $\hat{\theta}$, and is referred to as the **bootstrap (sampling) distribution** of $\hat{\theta}$.

Diagram describes the process of generating bootstrap statistics:



Bootstrap Estimation –Examples

Suppose that we have a random sample $y_1, y_2, \ldots, y_n$ with $Y_i \sim \text{Exp}(\lambda)$.

We wish to make inferences about λ.

From CLT (asymptotically) $\bar{Y} \sim N(\lambda^{-1}, (n\lambda^2)^{-1})$

We can use this to estimate quantities of interest (e.g. for confidence intervals or tests for λ).

What if assumptions or asymptotic results may not hold - e.g.

- small sample?
- and/or unknown/non-symmetric distribution?

Bootstrap allows us to avoid making assumptions about the sampling distribution of a statistic of interest, by instead **forming an empirical sampling distribution of the statistic**.

This is generally achieved with resampling from the available sample.

(a) Non-parametric (full) bootstrap

We **resample with replacement from the original sample**:

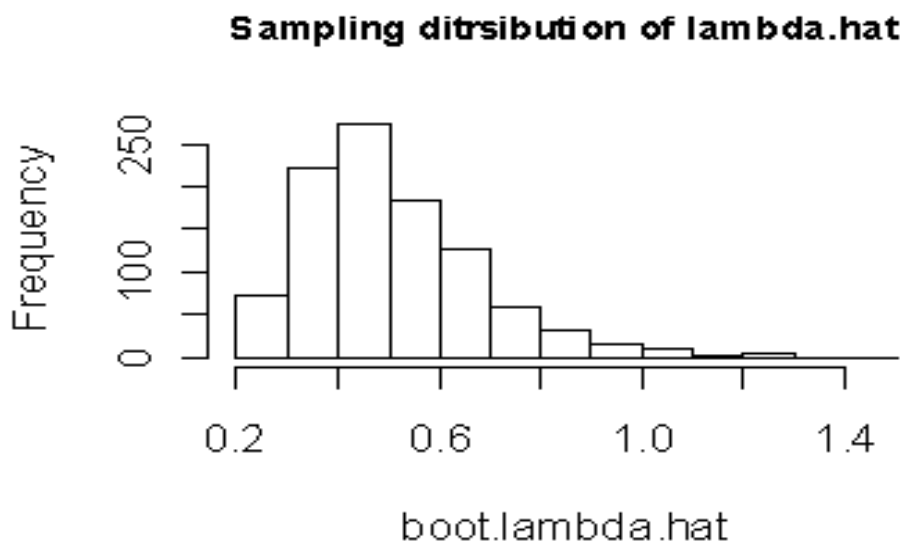
for j in (1 to B)

1. obtain j^{th} sample $y_1^*, y_2^*, \ldots, y_n^*$ by resampling with replacement from $y_1, y_2, \ldots, y_n$
2. compute $\bar{y}_j^*$ and obtain $\hat{\lambda}_j^* = 1/\bar{y}_j^*$

With B sufficiently large, the output set of estimates $(\hat{\lambda}_1^*, \ldots, \hat{\lambda}_B^*)$ can be used to **provide an empirical sampling distribution of** $\hat{\lambda}$

Schematically, this can be depicted as

$$y_1, y_2, \ldots, y_n \rightarrow \left.\begin{array}{ll} \text{sample 1}: & (y_1^*, y_2^*, \ldots, y_n^*) \rightarrow \hat{\lambda}_1^* \\ \text{sample 2}: & (y_1^*, y_2^*, \ldots, y_n^*) \rightarrow \hat{\lambda}_2^* \\ \vdots & \\ \text{sample } B: & (y_1^*, y_2^*, \ldots, y_n^*) \rightarrow \hat{\lambda}_B^* \end{array}\right\} \rightarrow \begin{array}{c}\text{empirical} \\ \text{distn.} \\ \text{of } \hat{\lambda}\end{array} \qquad \text{e.g.}$$



Hence we can:

- estimate the mean of $\widehat{\lambda}$ by

$$\hat{E}(\hat{\lambda}) = \frac{1}{B}\sum_{j=1}^{B}\hat{\lambda}_j^*$$

- estimate its median, using the empirical quantiles of the bootstrap values λ_j^* , i.e.

$$\widehat{\text{median}}(\hat{\lambda}) = Q_e(0.5)$$

- estimate its variance:

$$\widehat{\text{Var}}(\hat{\lambda}) = \frac{1}{B-1}\left(\sum_{j=1}^{B}\left(\hat{\lambda}_j^*\right)^2 - \frac{1}{B}\left(\sum_{j=1}^{B}\hat{\lambda}_j^*\right)^2\right)$$

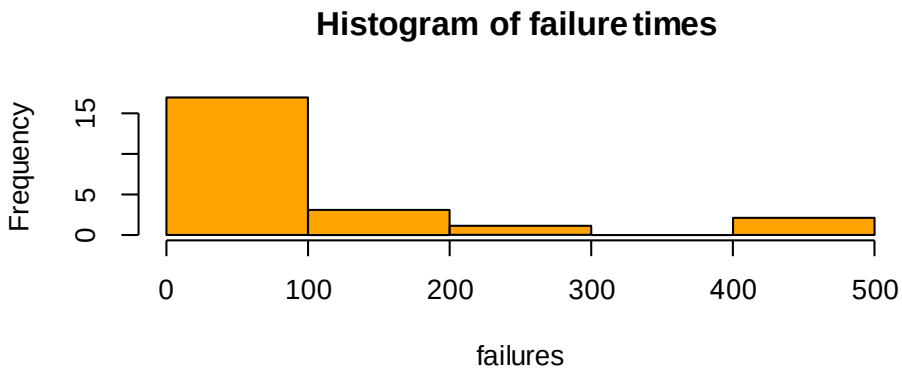
- Obtain $100(1-\alpha)\%$ confidence intervals:

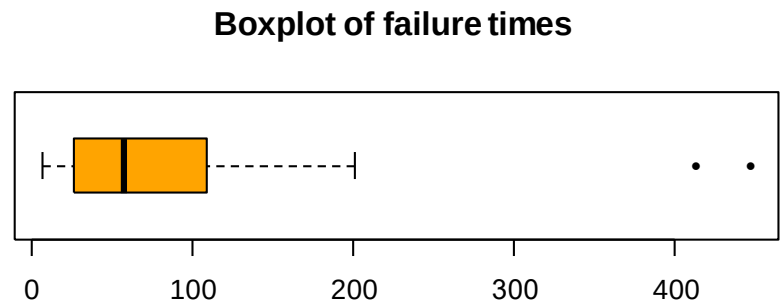
$$[Q_e(\alpha/2),\ Q_e(1-\alpha/2)]$$

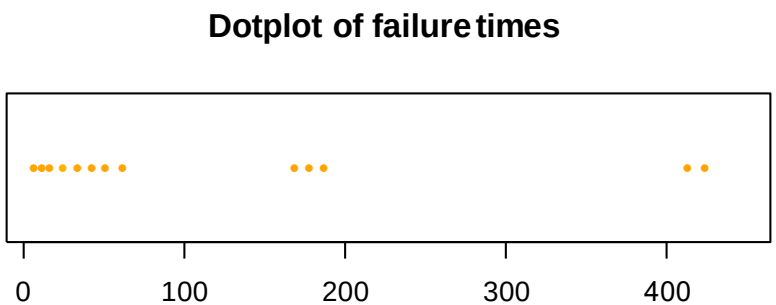
- Compute p-values for hypothesis tests (see later) etc.

Example 4.1:
The numbers of operating hours between successive failures of the air-conditioning equipment aboard a Boeing 720 aircraft were as follows:

413	14	58	37	100	65	9	169	447	184	36	201
118	34	31	18	18	67	57	62	7	22	34	







Data distribution has long tail to the right.

We estimate the population median, m, by the sample median, $\hat{m} = 57$.

How **accurate** is this estimate?

Can we use the estimator provided by the sample median to construct a CI for m?

- Asymptotic normality *doubtful* ($n = 23$)
- Use bootstrap!

We would like the sampling distribution of $\hat{m}$ ...

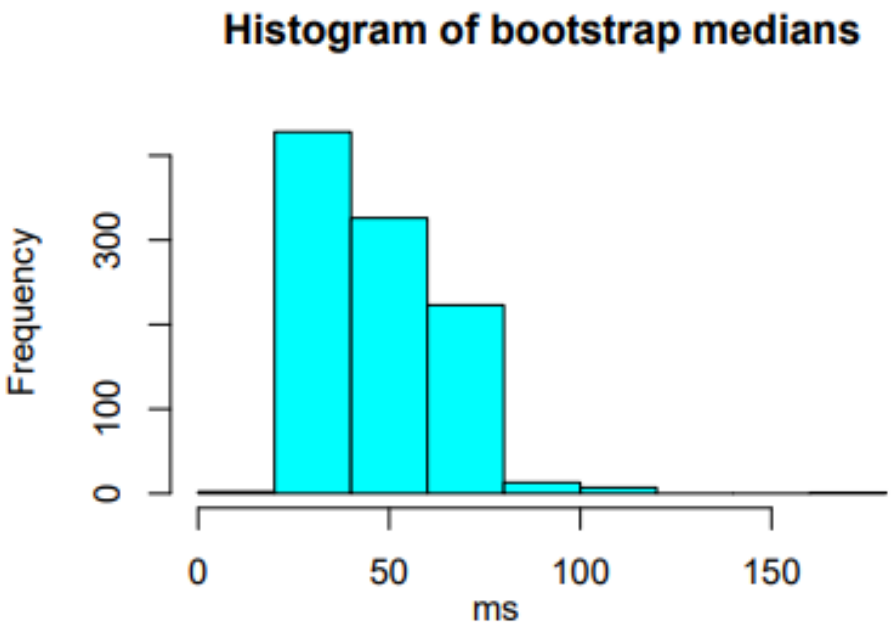
... but this requires knowledge of the (unknown) underlying population distribution

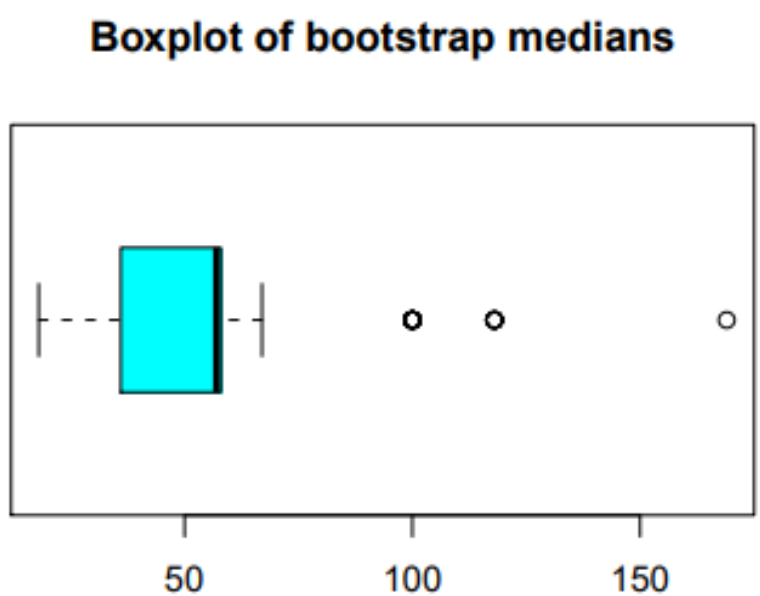
Use instead the empirical distribution of the data (obtained with bootstrap!)

Set up an R vector, median.b, of size B=1000 and use it to store the results of 1000 simulations of $\hat{m}$.

```
B = 1000
median.b = numeric(B)
for(i in 1:B) median.b[i] = median(sample(failures, size=23, replace=TRUE))
```

```
> summary(median.b)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  18.00   36.00   57.00   50.54   62.00  169.00
```





Recall that each component of median.b is the median of a random sample of size 23. Hence the variability in median.b is much less than the variability in failures itself.

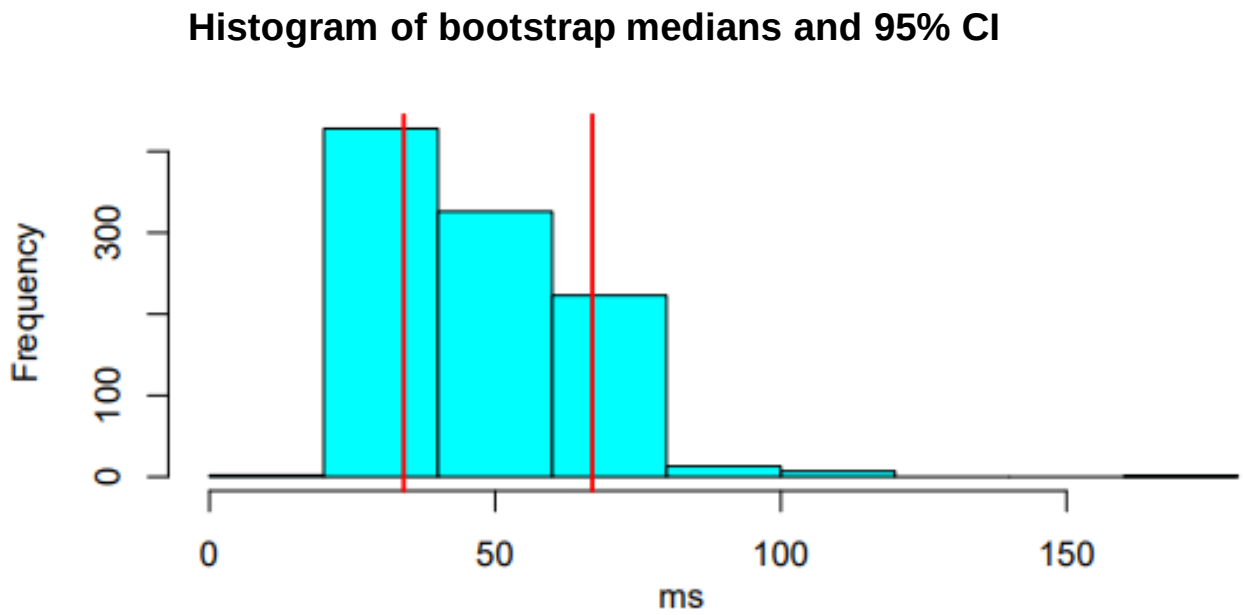
A reasonable 95% CI for the original population median m is given by the so-called **percentile bootstrap CI**:

$$(Q_e(0.025), Q_e(0.975))$$

where Q_e is the empirical quantile function of the bootstrap simulations median.b of $\hat{m}$.

```
> quantile( median.b, c(0.025,0.975) )
```

so that (34, 67) is a reasonable 95% confidence interval for m.



(b) Parametric bootstrap

If we are *prepared to assume* that the sample is considered to come from Exp(λ), then start by obtaining an estimate of λ : $\hat{\lambda} = 1/\bar{y}$

Then **use the Exp($\hat{\lambda}$) distribution to draw the bootstrap samples.**

Once the bootstrap samples are available, we proceed as with the non-parametric method before.

Algorithm:

for j in (1 to B)

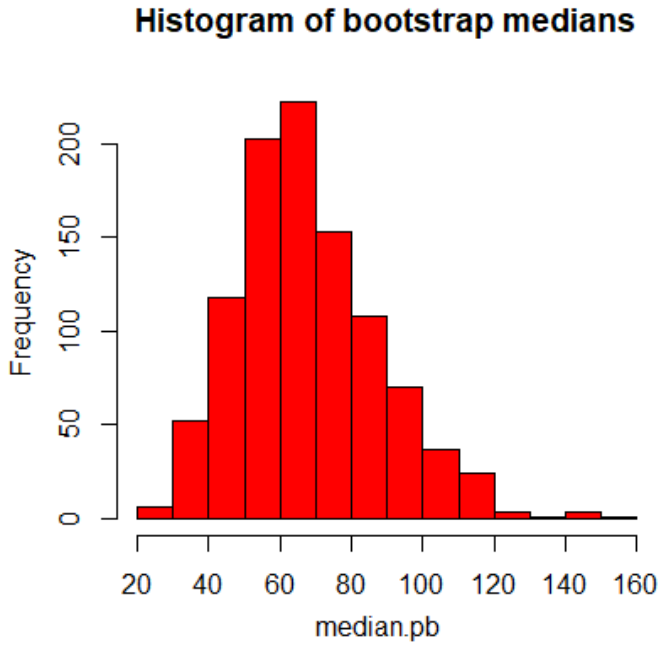
1. obtain new (j^{th}) sample $y_1^*, y_2^*, \ldots, y_n^* \sim \text{Exp}(\hat{\lambda})$
2. compute $\bar{y}_j^*$ and obtain $\hat{\lambda}_j^* = 1/\bar{y}_j^*$

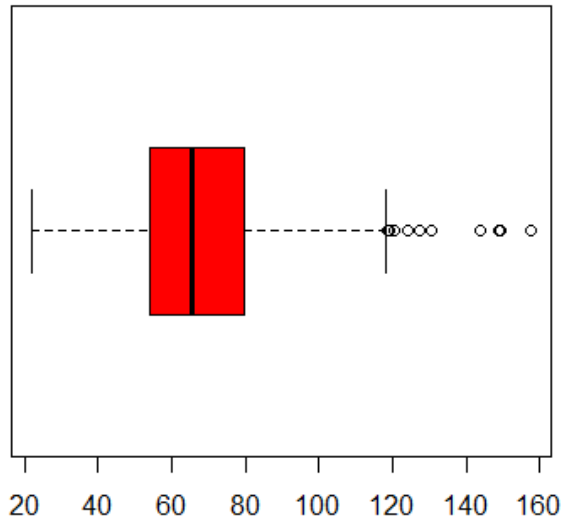
Various inferences can then be made as before.

Set up an R vector, median.pb, of size B=1000 and use it to store the results of 1000 simulations of $\hat{m}$.

```
B = 1000
lambdahat= 1/mean(failures)
median.pb = numeric(B)
for(i in 1:B) median.pb[i] = median(rexp(23, rate=lambdahat))
```

```
> summary(median.pb)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  18.24   46.05   57.65   59.05   70.95  118.04
```

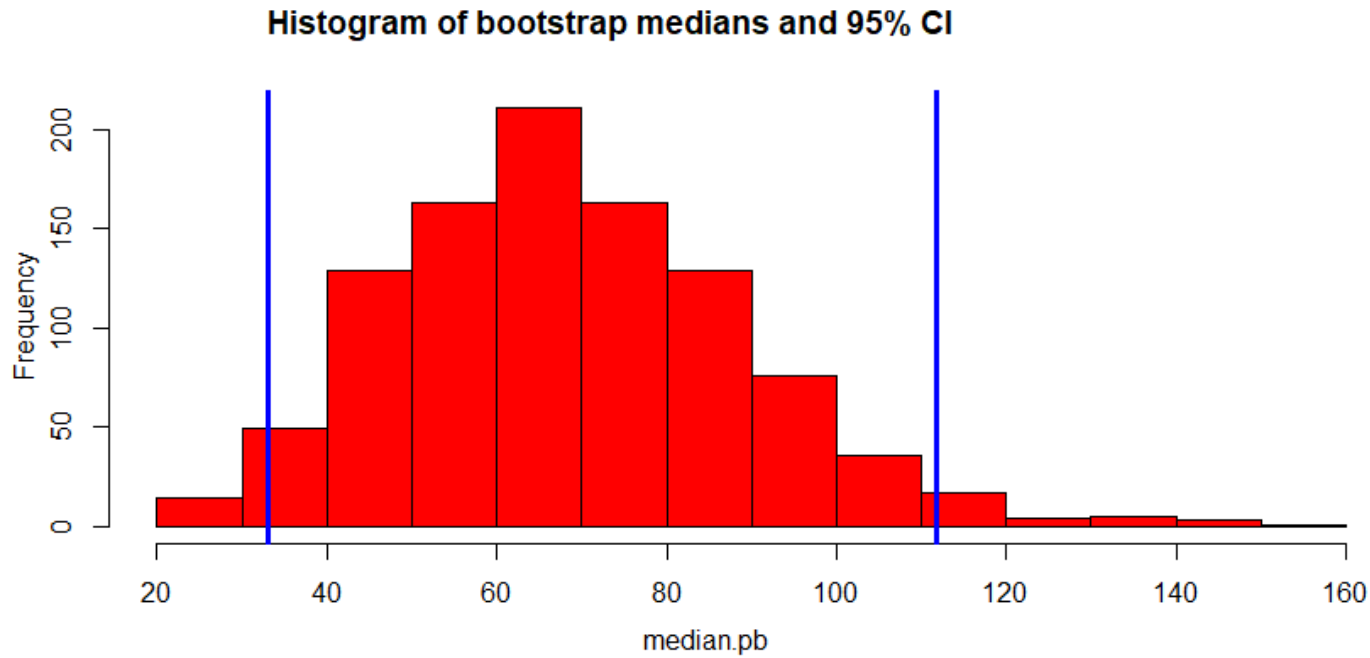




A reasonable 95% bootstrap CI for population median m using parametric bootstrap method is given by:

```
> quantile(median.pb,c(0.025,0.975))
     2.5%     97.5%
33.07168 111.81720
```

so that $(33.07,\ 111.82)$ is a reasonable 95% confidence interval for m.



Appendix

Calculating maximum-likelihood estimation (MLE) of the exponential distribution.

The probability density function of the exponential distribution is defined as

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Its likelihood function is

$$\mathcal{L}(\lambda, x_1, \ldots, x_n) = \prod_{i=1}^{n} f(x_i, \lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x} = \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}$$

To calculate the maximum likelihood estimator solved the equation for λ.

$$\frac{d \ln(\mathcal{L}(\lambda, x_1, \ldots, x_n))}{d\lambda} = 0$$

So,

$$\begin{aligned} \frac{d \ln(\mathcal{L}(\lambda, x_1, \ldots, x_n))}{d\lambda} &= \frac{d \ln\left(\lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}\right)}{d\lambda} \\ &= \frac{d \ln(n \ln(\lambda) - \lambda \sum_{i=1}^{n} x_i)}{d\lambda} \\ &= \frac{n}{\lambda} - \sum_{i=1}^{n} x_i \end{aligned}$$

Finally we get $\hat{\lambda} = \dfrac{n}{\sum_{i=1}^{n} x_i} = \dfrac{1}{\bar{x}}$ #